

Perspective Correction of Oblique Weightlifting Videos for Biomechanical Analysis

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Abstract

Video-based biomechanical analysis of weightlifting movements enables objective assessment of barbell trajectory and joint kinematics, but its accuracy depends heavily on camera placement and athlete orientation. In practical training environments, cameras may not be aligned perpendicular to the lifter's sagittal plane, while the lifter may also adopt a non-orthogonal stance or rotate during the lift. These effects can introduce view-dependent distortion in the measured barbell trajectory and 2D joint kinematics and may be difficult to distinguish from a single image.

This project proposes a digital image processing pipeline for detecting and compensating for view-dependent geometric distortion in monocular weightlifting videos without requiring prior knowledge of whether the distortion originates primarily from camera placement or athlete orientation. The system estimates an apparent viewing/orientation measure using two complementary cues: geometric analysis of the barbell plate's elliptical projection and depth/orientation information obtained from MediaPipe 3D pose landmarks. These estimates are evaluated temporally to distinguish predominantly static viewing geometry from changes in athlete orientation during the lift. Confidence measures are incorporated to account for plate-detection errors, pose-estimation uncertainty, and occlusion.

For planar elements of the scene, such as the lifting platform and barbell trajectory reference plane, homography-based perspective rectification is investigated where appropriate. For the athlete, rather than applying a single planar homography to the body, the estimated 3D pose is used to characterize out-of-plane orientation and its effect on 2D joint-angle measurements. The corrected barbell trajectory and joint-angle measurements are then compared with measurements obtained from a near-orthogonal reference camera.

Image enhancement techniques, including CLAHE-based contrast enhancement, denoising, and motion-blur reduction, are applied as preprocessing steps to improve plate and pose detection under variable lighting and motion conditions. A subset of lifts is recorded simultaneously using oblique and near-orthogonal cameras to provide a reference for evaluating the proposed correction. The project aims to determine whether angle-aware geometric correction and image enhancement can improve the reliability of biomechanical analysis.

bility of barbell trajectory and 2D biomechanical measurements obtained from low-cost, non-ideal monocular recordings.